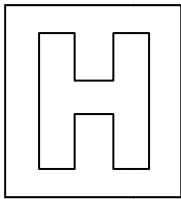


Candidate Name: _____

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2015 Preliminary Examination 2

Pre-University 3

H2 Biology**9648/01****Paper 1 Multiple Choice****25 September 2015****1 hour 15 min**Additional material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

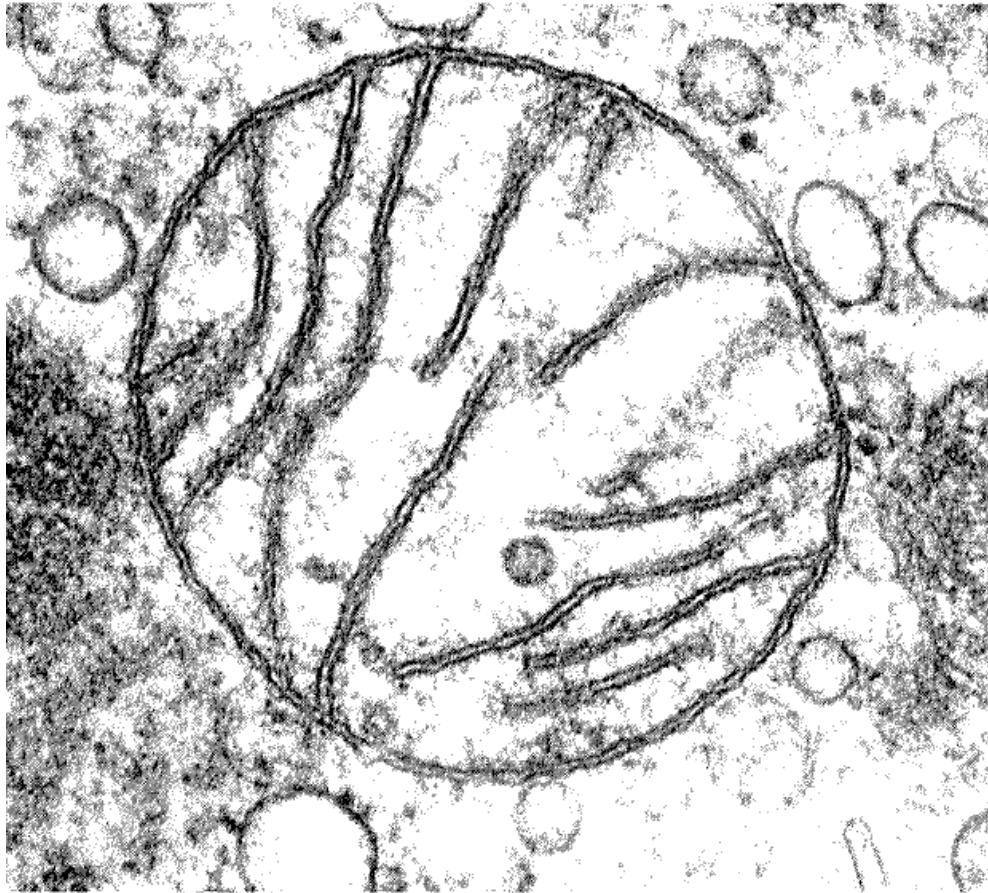
Write your name and Admission No. on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate

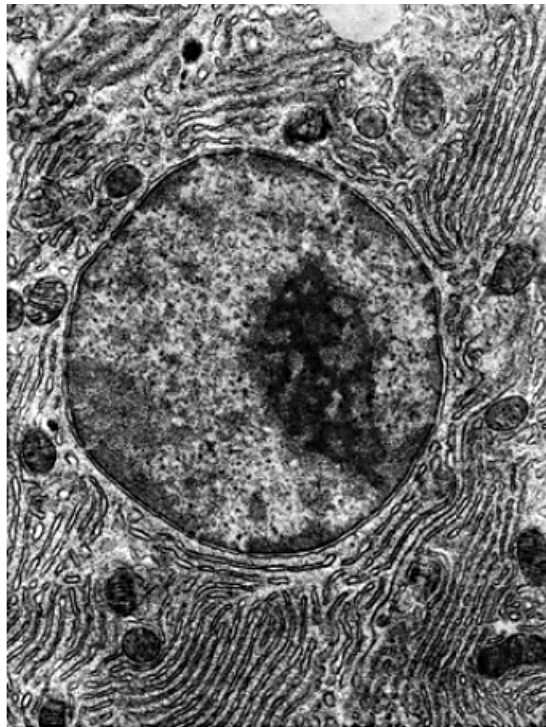
1. The transmission electron micrograph shows a transverse section of a cellular organelle. The magnification of the electron micrograph is $\times 100000$.



Which statement correctly describes this organelle?

- A** The cristae of a mitochondrion has a diameter of approximately 200 nm.
- B** The cristae of a mitochondrion has a diameter of approximately 20 nm.
- C** The thylakoids of a chloroplast has a diameter of approximately 200 nm.
- D** The thylakoids of a chloroplast has a diameter of approximately 20 nm.

2. The electron micrograph shows part of an animal cell.



What will be synthesised in large quantities in this cell?

- 1 ATP
 - 2 Glucose
 - 3 RNA
- A** 1 only
- B** 2 only
- C** 1 and 3 only
- D** All of the above
3. Membranous sacs containing products of metabolism are formed by the rough endoplasmic reticulum in cells.

Where are these products used?

- A** Inside and outside the cell
- B** Inside lysosomes only
- C** Inside the cell only
- D** Outside the cell only

4. Which of the statements about polysaccharides can be used to describe both starch and cellulose?

- 1 Every adjacent glucose molecules are rotated by 180°
- 2 Contains glycosidic bonds
- 3 Polymer of α -glucose

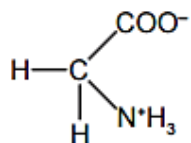
- A 1 only
- B 2 only
- C 1 and 3 only
- D 2 and 3 only

5. Some foods contain 'hydrogenated vegetable oils'. These are unsaturated fats that have been converted to saturated fats.

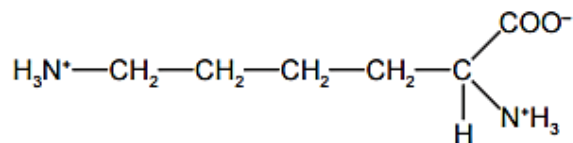
Which property of the fats will have changed?

- A Their hydrocarbon chains will pack together more closely.
- B Their solubility in water will increase.
- C They will have more double bonds in their molecules.
- D They will remain liquid at room temperature.

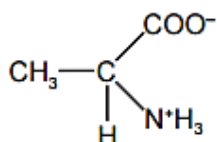
6. The diagram shows the structure of four amino acids in solution.



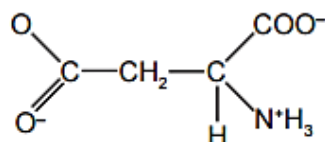
glycine



lysine



alanine



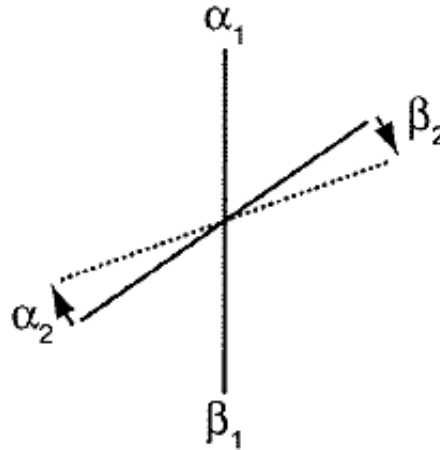
aspartate

Which of these four amino acids have an overall charge?

- A** alanine and aspartate
- B** alanine and glycine
- C** glycine and lysine
- D** aspartate and lysine

7. A molecule of haemoglobin is made up of two α polypeptide subunits (α_1 and α_2) and two β polypeptide subunits (β_1 and β_2). The relative positions of these subunits change when a deoxygenated haemoglobin molecule takes up oxygen.

The axis joining the α_2 and β_2 subunits rotates by about 15° in relation to the axis joining the α_1 and β_1 subunits, as shown in the diagram.



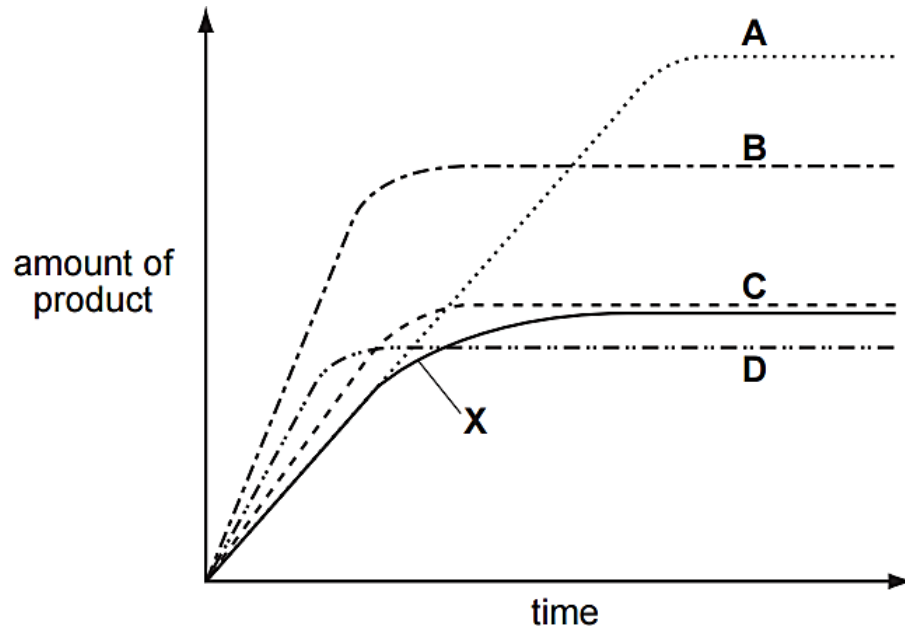
Which statements about this rotation are correct?

- 1 The points of contact between the four subunits are altered.
- 2 The rotation resulting from adding one oxygen molecule to one of the subunits makes it easier to add oxygen to the other subunits.
- 3 The rotation alters the quaternary structure of the molecule.
- 4 The rotation makes different amino acids available for binding oxygen.

- A** 1, 2 and 3
B 1, 2 and 4
C 1, 3 and 4
D 2, 3 and 4

8. The curve X shows the activity of an enzyme at 20 °C with small amount of substrate available. Curves **A**, **B**, **C** and **D** show the effect of different conditions on the activity of the enzyme.

Which curve shows the effect of increasing the temperature by 10°C and adding extra substrate?



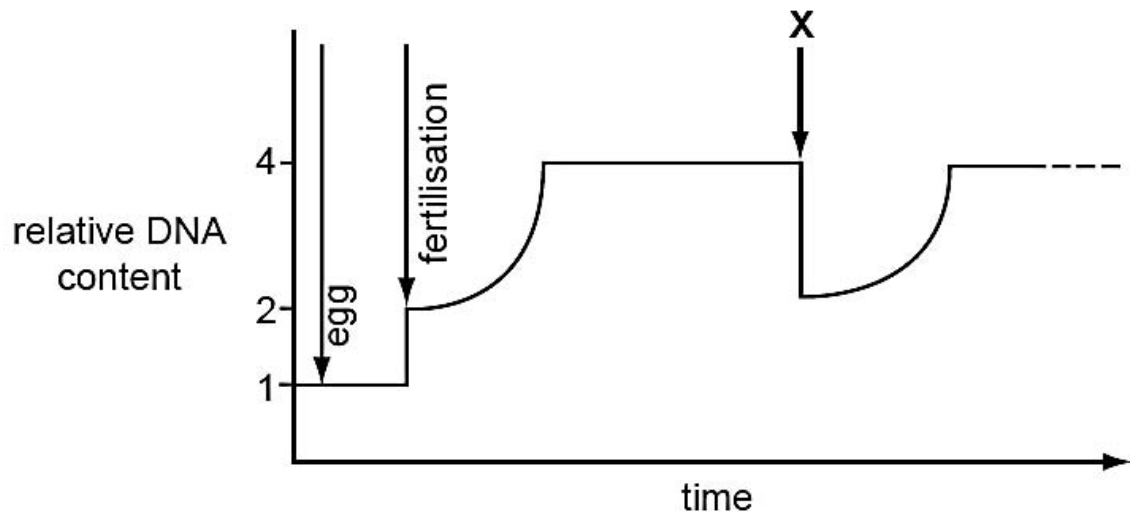
9. The fluidity of the cell surface membrane can be changed by a number of factors.

As the fluidity of cell surface membranes decreases, which process would be least affected?

- A** active transport
- B** diffusion
- C** endocytosis
- D** osmosis

- 10.** What supports the view that a membrane protein is involved in active transport?
- A** It allows movement of molecules across a membrane if concentration gradient exists.
 - B** It can only function if mitochondria are supplied with sufficient oxygen.
 - C** It has a tertiary structure with a binding site with a specific shape.
 - D** It is found in the cell surface membranes and the mitochondrial membranes.
- 11.** Which are the features of nuclear division by mitosis?
- 1 forms cells of equal size to the parent cell
 - 2 forms genetically identical cells
 - 3 semi-conservative replication of DNA
- A** 2 only
 - B** 1 and 2 only
 - C** 2 and 3 only
 - D** 1, 2 and 3

12. The graph represents the changes in the quantity of DNA present in one nucleus at different stages in the life cycle.

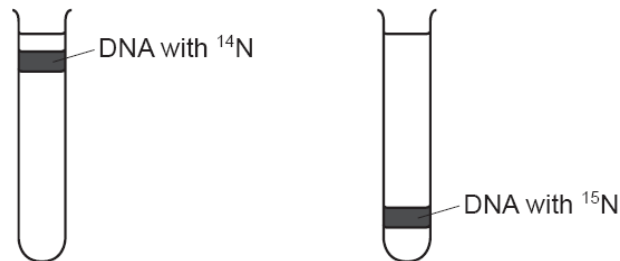


Which stage takes place at X?

- A Interphase
- B Metaphase
- C Prophase
- D Telophase

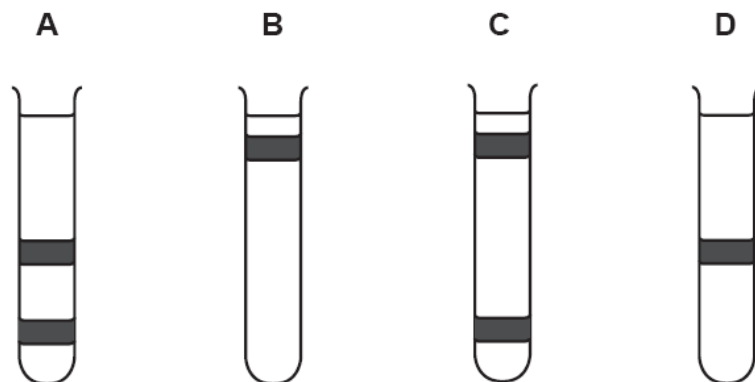
13. Two sets of bacteria were grown using different types of nitrogen-containing growth media. One set was grown in a medium containing the 'heavy' isotope of nitrogen, ^{15}N , until all DNA was labelled. The other set were grown in a medium containing the 'light' isotope of nitrogen, ^{14}N , until all DNA was labelled.

The DNA from each set of bacteria was extracted and centrifuged. The diagram shows the position in the centrifuge tubes of this DNA.



Bacteria with ^{15}N labelled DNA were transferred to a medium containing ^{14}N and allowed to reproduce once. The DNA of the new generation of bacteria was extracted and centrifuged.

Which tube shows the position of DNA from this new generation of bacteria?



14. In a DNA molecule, the base sequence AGT codes for the amino acid serine.

What is the base sequence of the anti-codon on the tRNA to which serine becomes attached?

- A AGU
- B GAU
- C TCA
- D UCA

15. One individual, organism **P**, in a sexually-reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

Normal protein	Variant protein
-gly-lys-lys-gly-glu-	-gly-lys-gly-gly-glu-

What could account for the presence of the normal protein and the variant protein in **P**?

- 1 The sequence of amino acids was altered in a cell in the heart muscle of **P**'s mother.
 - 2 The sequence of amino acids was altered in a cell in the ovaries of **P**'s mother.
 - 3 The sequence of bases was altered in the DNA in a cell in the heart muscle of **P**.
 - 4 The sequence of bases was altered in the DNA in a cell in an ovary of **P**.
 - 5 The sequence of bases was altered in the DNA in a cell in a testis of **P**'s grandfather.
- A** 1 and 3 only
B 3 and 5 only
C 1, 2 and 4 only
D 2, 4 and 5 only

16. The table shows a comparison of some aspects of the genomes and protein-coding genes of the prokaryote bacterium *Escherichia coli* and the eukaryote fungus *Saccharomyces cerevisiae*.

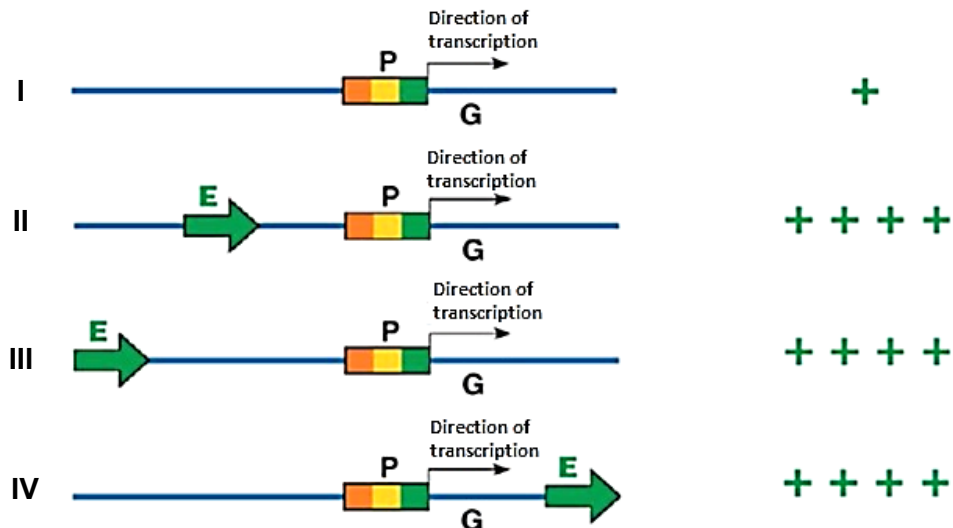
	<i>E. coli</i>	<i>S. cerevisiae</i>
genome length / base pairs	4 640 000	12 068 000
number of protein-coding genes	4 300	5 800

What could **not** help account for the differences in the number of protein-coding genes?

- A the large number of proteins involved in transcription in eukaryotes
 - B the presence of membrane-bound organelles in eukaryotes
 - C the use of histones to package DNA in eukaryotes
 - D the presence of introns in the DNA of eukaryotes
17. What describes common features of the telomere and the centromere?
- A Both areas have proteins associated with them.
 - B Both are made up of DNA with a high incidence of adenine and thymine.
 - C Both DNA sequences are conserved throughout the life of the organism.
 - D During interphase, DNA replication takes place simultaneously in both regions.

18. Recombinant DNA techniques can be used to alter the locations of control elements within DNA sequences, so as to study the effects of these changes on the levels of transcription. The diagram shows the various structures of transcription units and the corresponding relative levels of transcription.

The promoter, enhancer and coding sequences are represented by letters P, E and G respectively. The number of symbol '+' indicates the relative levels of transcription.



With reference to the diagram, which of these statements is a valid conclusion?

- A** The relative distance between promoter and enhancer has no effect on the rate of transcription.
- B** The rate of transcription is increased when the enhancer is located upstream of the promoter.
- C** An enhancer is required for transcription.
- D** A promoter bound with RNA polymerase and general transcription factors result in basal rate of transcription.

19. Which statement about cancer is incorrect?

- A** Mutations in proto-oncogenes may not necessarily result in excessive cell proliferation.
- B** Two recessive alleles for a tumour suppressor gene must be present.
- C** The risk of cancer is higher when *p53* gene is translocated to another chromosome having a strong promoter.
- D** The growth of new blood vessels is a fundamental step in the development of a malignant tumour.

20. Which of the following correctly describes a prokaryotic genome?

	Presence of homologous chromosomes	Presence of histone proteins	Presence of DNA in organelles	Presence of plasmids
A	✓	✓	✓	x
B	x	✓	✓	x
C	✓	x	x	✓
D	x	x	x	✓

21. A mutant strain of *E. coli* has been isolated in which the *lac* operon is not expressed in the presence of lactose. To understand the nature of this defect, this parent strain was mated so that it now contains an F' factor containing a normal *lac* operon. The parent and mated strain with regard to their β -galactosidase activities in the presence and absence of lactose was compared. The following results were obtained:

Strain	Addition of lactose	Amount of β -galactosidase (percentage of parent strain)
Parent	No	0
Parent	Yes	0
Mated	No	0
Mated	Yes	100

With respect to the results shown in table, which part of the operon most likely has the defect/mutation?

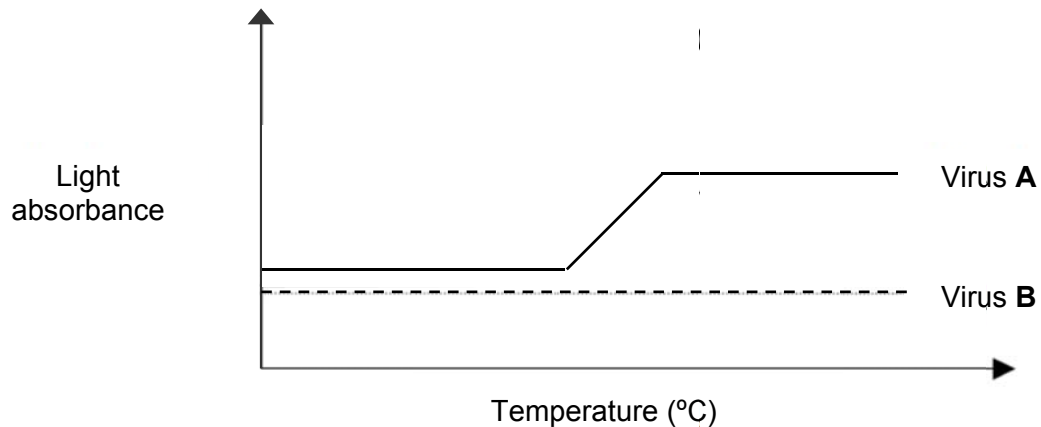
- A *lacA*
 - B *lacI*
 - C Promoter
 - D Operator
22. Viruses can be considered to be living organisms.

What is **not** an argument that in any way supports the above statement?

- A All viruses carry coded information that can result in the production of polypeptides.
- B Molecules located on the surface of a virus can bind to complementary-shaped receptors on host cells.
- C The genetic material of a virus can be replicated and its genes expressed.
- D Some viruses contain enzymes which may be used in their reproductive cycles.

23. Two new viruses **A** and **B**, which infect eukaryotic cells, have been identified.

In one experiment, the nucleic acid from each virus is isolated and analyzed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The behaviour of the nucleic acid from each virus is shown in the graph.



In a second experiment, it is found that treatment with reverse transcriptase inhibitors blocks the ability of virus **B** to infect cells. In contrast, reverse transcriptase inhibitors have no effect on the ability of virus **A** to infect cells. In a third experiment, it was revealed that DNA synthesis inhibitors block infection by virus **A**.

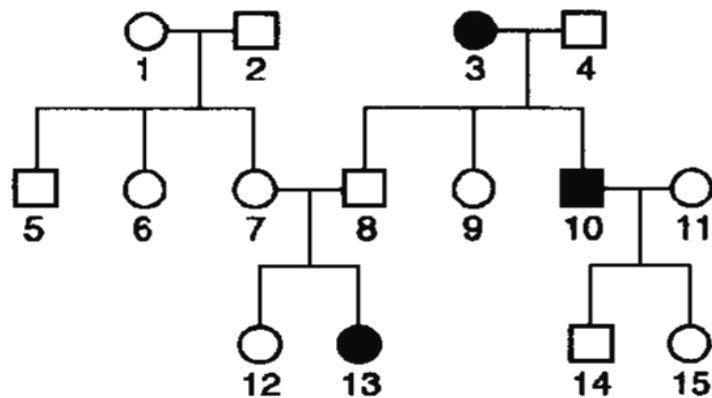
Which of the following conclusions can be drawn from both the experiments?

- A** The genome of virus **A** is single-stranded RNA and that of virus **B** is double-stranded DNA.
- B** The genome of virus **A** is double-stranded DNA and that of virus **B** is single-stranded RNA.
- C** The genome of virus **A** is double-stranded RNA and that of virus **B** is single-stranded DNA.
- D** The genome of virus **A** is double-stranded DNA and that of virus **B** is double-stranded RNA.

24. Which of the following statements is false?

- A Continuous variation shows a normal distribution and is influenced by both genetic and environmental factors.
- B Continuous variation is controlled by many genes and the traits are usually poorly defined with varying degree of gradation.
- C Discontinuous variation shows traits that follow discrete distribution and is mostly influenced by environmental factors.
- D Discontinuous variation shows traits that are controlled by one or two genes and is relatively unaffected by environmental factors.

25. The family tree shows the inheritance of a condition caused by the change in nucleotide sequence of gene A.



key

- | | |
|-----------------|-------------------|
| ○ normal female | ● affected female |
| □ normal male | ■ affected male |

What is the probability that the first child is an affected girl when individual 7 mated with an affected male instead?

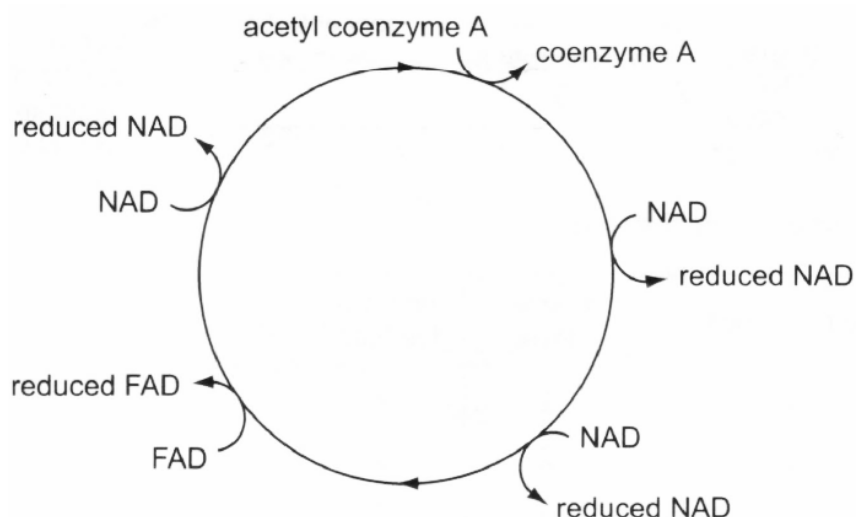
- A 0.75
- B 0.50
- C 0.25
- D 0

26. If a non-disjunction of chromosomes occurred in the first meiotic division, and one of the resulting gamete fuses with a normal gamete, which of the following zygote(s) may possibly be formed?

- I Normal zygote
- II Monosomic zygote
- III Trisomic zygote

- A III only
- B II and III only
- C I and II only
- D All of the above

27. The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.

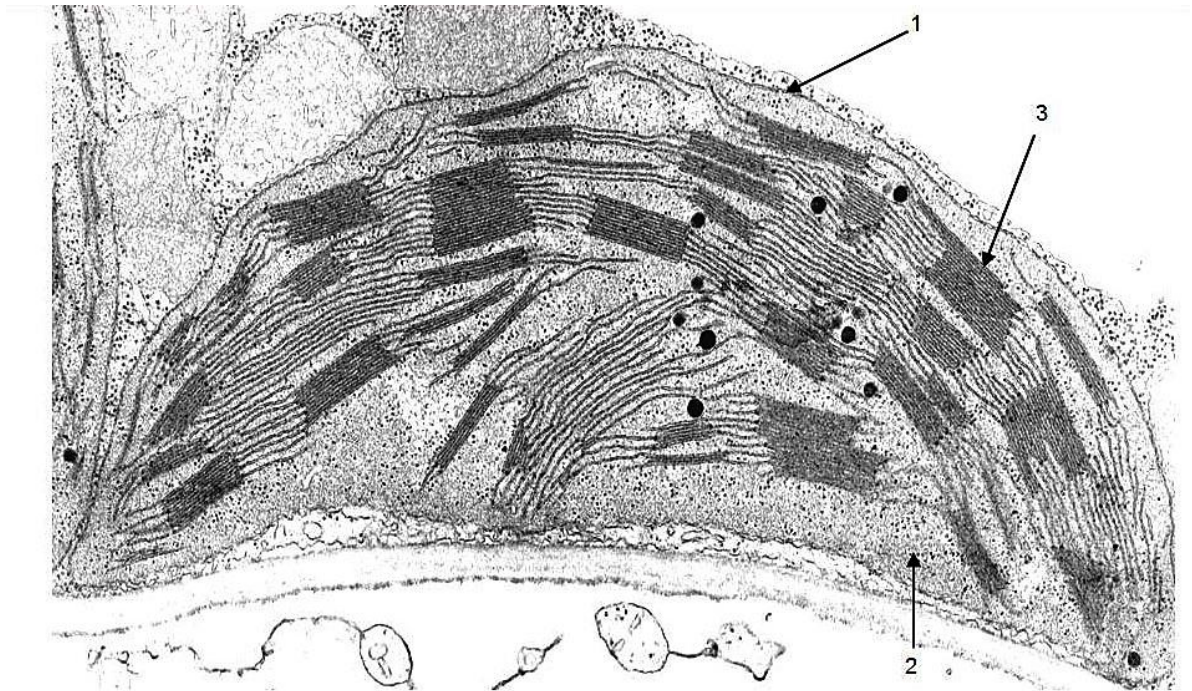


The average yield of ATP, in oxidative phosphorylation, is 3 molecules from each molecule of reduced NAD and 2 molecules from each molecule of reduced FAD.

What is the average yield of ATP from the hydrogen carriers reduced in the Krebs cycle from one molecule of glucose?

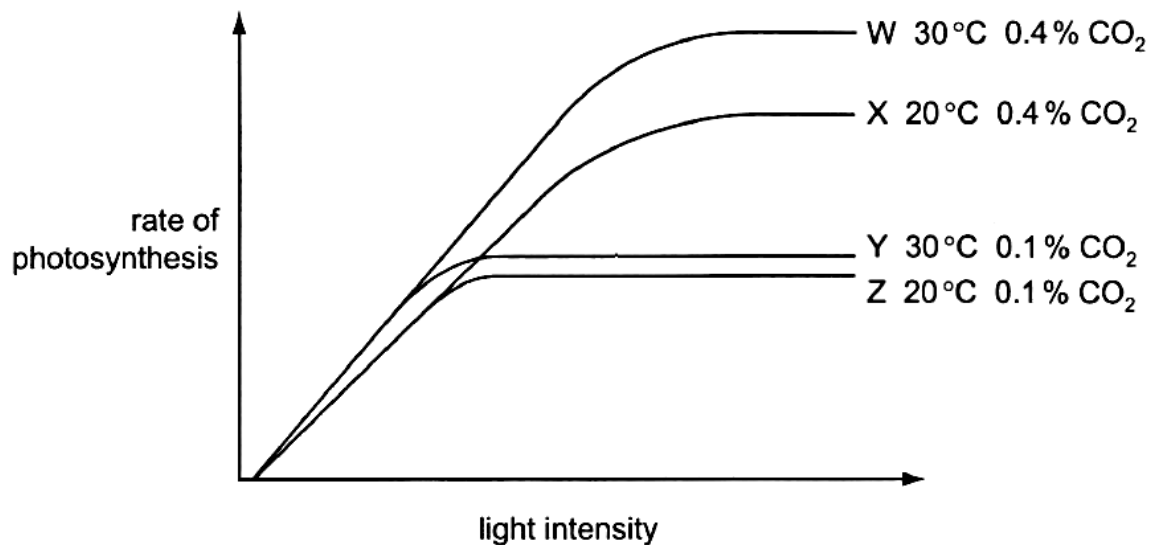
- A 11
- B 22
- C 34
- D 36

28. Which of the following is correctly associated with the function or characteristic of the labelled structure?



	Contain ATP synthase	Site where sugar is made	A double membrane structure
A	1	3	2
B	2	1	3
C	3	1	2
D	3	2	1

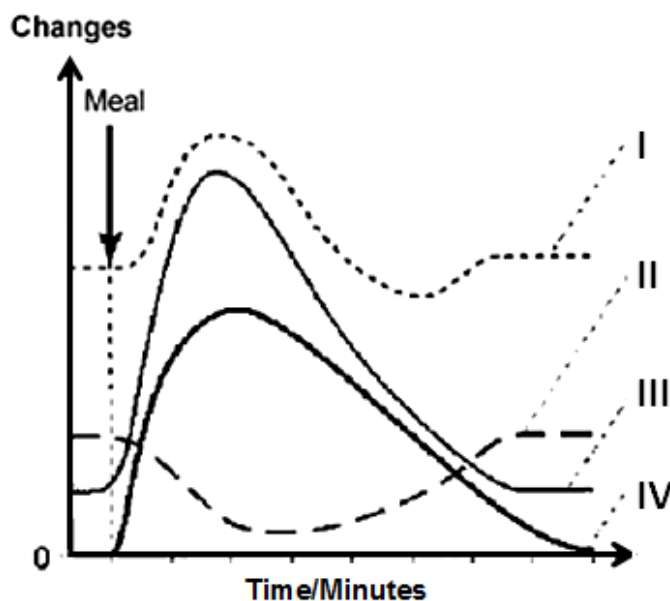
29. The diagram shows the results of an investigation into the effect of changing light intensity on the rate of photosynthesis at two different carbon dioxide concentrations and two different temperatures.



Which factor is limiting the rate of photosynthesis shown in curve **X** at high light intensities and which curve supports this?

- A** Carbon dioxide concentration, curve **Y** as a decrease in carbon dioxide concentration decreases rate.
- B** Carbon dioxide concentration, curve **Z** as rate becomes constant at lower light intensities.
- C** Temperature, curve **W** as an increase in temperature increases rate.
- D** Temperature, curve **Z** as rate becomes constant at lower light intensities.

30. The figure below shows changes in the levels of insulin, blood glucose, gluconeogenesis, and glucose taken up by muscles cells over time after a meal.



Which of the following options correctly identifies and explains the levels of insulin, blood glucose, gluconeogenesis and glucose uptake by muscles cells to their corresponding graphs?

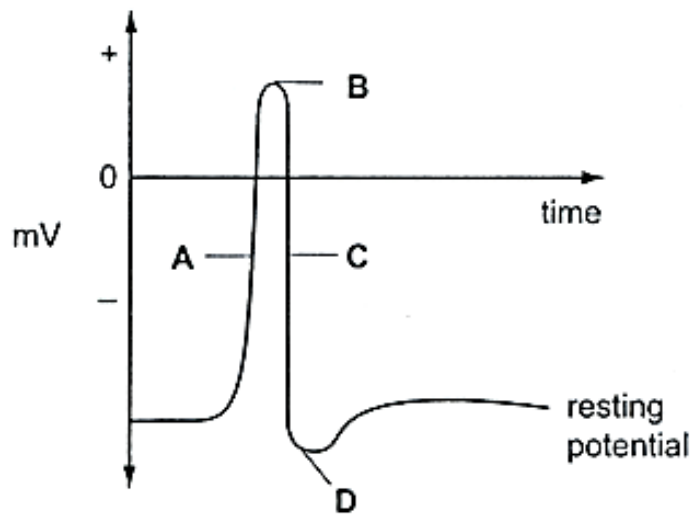
	I	II	III	IV
A	Insulin level increases due to increased glucose binding to receptors in the β cells	Glucose uptake by muscles cells increases after meal to decrease blood glucose level	Blood glucose level is maintained at a constant level due to negative feedback	Rate of gluconeogenesis decreases due to effects of insulin on insulin receptors in liver cells
B	Rate of gluconeogenesis decreases due to effects of insulin on insulin receptors in liver cells	Insulin level increases due to increased glucose binding to receptors in the β cells	Glucose uptake by muscles cells increases after meal to decrease blood glucose level	Blood glucose level is maintained at a constant level due to negative feedback
C	Blood glucose level is maintained at a constant level due to negative feedback	Rate of gluconeogenesis decreases due to effects of insulin on insulin receptors in liver cells	Insulin level increases due to increased glucose binding to receptors in the β cells	Glucose uptake by muscles cells increases after meal to decrease blood glucose level
D	Glucose uptake by muscles cells increases after meal to decrease blood glucose level	Blood glucose level is maintained at a constant level due to negative feedback	Rate of gluconeogenesis decreases due to effects of insulin on insulin receptors in liver cells	Insulin level increases due to increased glucose binding to receptors in the β cells

31. Which of the following process(s) leads to signal amplification in the cell?

- I Binding of multiple signal molecules to the membrane bound receptor
- II Phosphorylation of the first kinase in a phosphorylation cascade
- III Branching of different signaling pathways that produces multiple cellular responses
- IV Activation of numerous intracellular receptors that acts as transcription factors in gene expression

- A II only
- B II and III only
- C I and IV only
- D All of the above

32. The diagram shows an action potential.



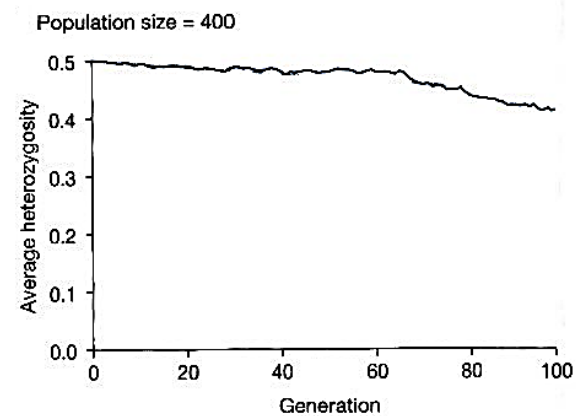
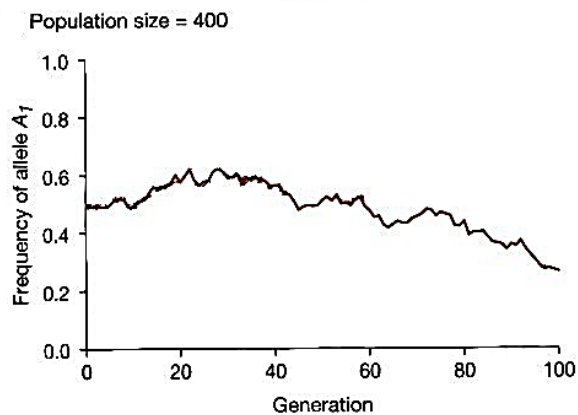
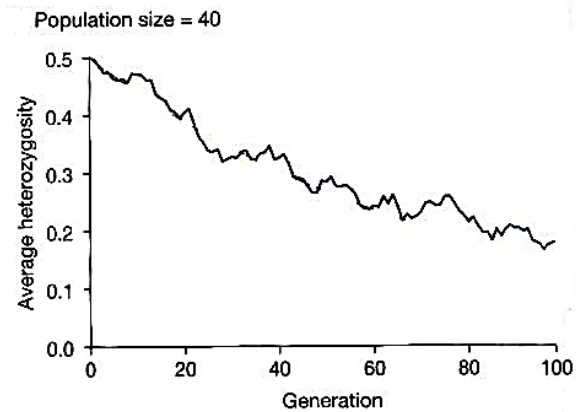
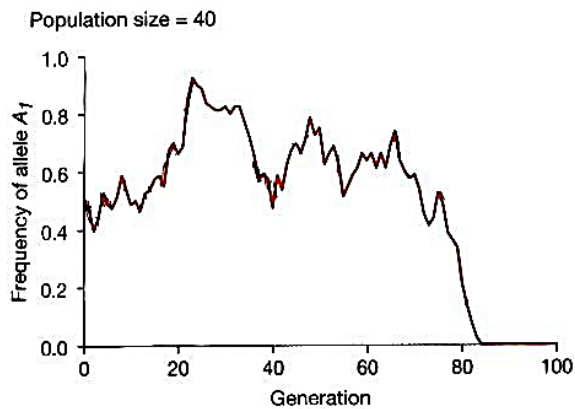
Which of the following statements is incorrect?

- A At point A, the axon membrane is most permeable to sodium ions.
- B At point B, the peak of the action potential can be reduced by decreasing the external Na^+ concentration.
- C At point C, the voltage-gated sodium channels are closed and potassium channels are open.
- D At point D, sodium conductance changes more slowly than does potassium conductance.

33. Which taxon is correctly described?

	Taxon	Size of the group	Number of similarities between members of the group	Ancestry of members of group
A	Family	Large	Few	recently shared a common ancestor
B	Species	Large	Few	distantly share a common ancestor
C	Family	Small	Many	distantly share a common ancestor
D	Species	Small	Many	recently shared a common ancestor

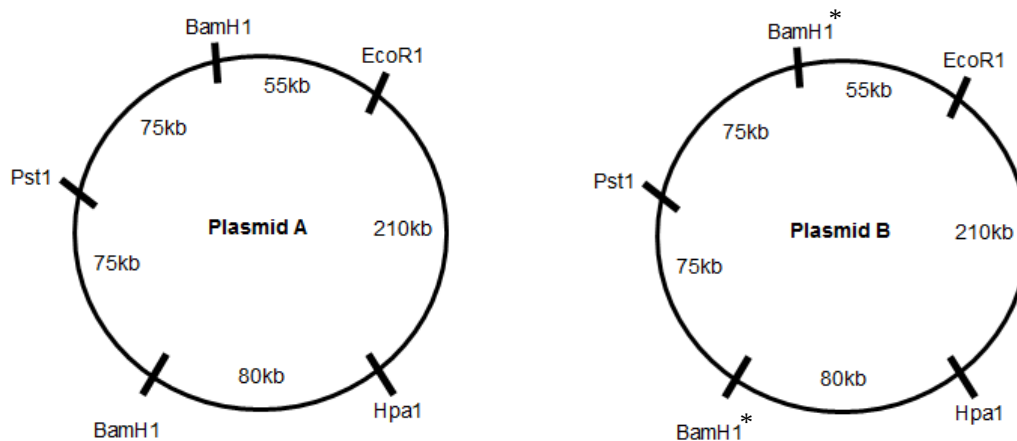
34. The four graphs represent simulations of genetic drift occurring in fruit fly populations of two different sizes, 40 and 400.



Which of the following about genetic drift can be deduced from the graphs?

- A** It leads to non-random fixation of alleles in the population.
- B** There is loss of heterozygosity in the population regardless of population size.
- C** It is a more powerful force of evolution in small populations.
- D** Given sufficient time, genetic drift can produce drastic changes in allele frequencies in large populations.

35. Figure below shows plasmid **A** and **B** as well as the respective restriction sites of various restriction enzymes. In plasmid B, mutations were deliberately introduced and affected restriction sites were denoted by *. The lengths of the plasmid DNA between consecutive restriction sites are also indicated.



A laboratory technician carried out an experiment by adding different restriction enzymes to the plasmids. Each tube was then left to incubate at 37°C for about 15 minutes. The experimental set-up and the results are shown in table below.

Tube number	Components	Fragment sizes / kb
1	Plasmid A + 2 restriction enzymes	80, 150, 265
2	Plasmid B + 2 restriction enzymes	155, 340

Which were the enzymes added to tubes 1 and 2?

	Tube 1	Tube 2
A	HpaI and EcoRI	HpaI and EcoRI
B	BamHI and HpaI	PstI and EcoRI
C	BamHI and EcoRI	HpaI and BamHI
D	BamHI and HpaI	HpaI and PstI

36. Seven sets of skeletons were excavated from an ancient Egyptian grave. Among them, there were four adults and three children. To establish the relationship between these seven individuals, small amount of DNA were isolated from these skeletons and polymerase chain reaction (PCR) was carried out. Genetic fingerprinting was then carried out on these DNA.

The chart shows some of the results obtained from the genetic fingerprinting of the skeletons.

Child 1	Child 2	Child 3	Adult 1	Adult 2	Adult 3	Adult 4
		—			—	
—	—				—	—
—	—	—	—		—	—
—	—	—	—	—		—
—	—	—	—	—		—
—	—	—	—	—	—	
—	—	—	—	—	—	
—	—	—	—	—	—	
				—		—
—		—	—		—	
	—	—			—	

Assuming that all the children had the same parents, which of the following show the children's parents?

- A Adult 1 and Adult 2
- B Adult 1 and Adult 3
- C Adult 2 and Adult 3
- D Adult 2 and Adult 4

37. What is the key reason for using a greater range of probes and restriction enzymes in DNA fingerprinting?
- A It permits more regions of the DNA to be analysed so as to reduce the possibility that two individuals' DNA would produce the same banding pattern.
 - B It increases the likelihood that one of the probes will bind to the polymorphic region of the DNA after the latter is cut by the restriction enzymes.
 - C It is necessary for the creation of unique DNA fingerprints from two individuals' DNA that are significantly different in sequence.
 - D It allows all the DNA bands produced via restriction enzymes cutting to be detected so as to give a more accurate DNA fingerprint.
38. Which uses of the information from the human genome project are generally considered to be unethical?
- I genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
 - II cytologists developing tests for only some defective genes
 - III an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases
 - IV doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
 - V genetic councillors giving specific lifestyle information only to people genetically predisposed to risks
 - VI parents choosing embryos for implantation only after prenatal tests for acceptable genes
- A I and II
 - B I and V
 - C III and IV
 - D III and VI

- 39.** It has been found that stem cells transferred from the intestinal lining to the bone marrow produce all of the different types of blood cell instead of intestinal cells.

Which statement best explains this?

- A** All stem cells are totipotent.
 - B** Environmental factors change the expression of specific genes.
 - C** Specific genes are destroyed by endonucleases.
 - D** Specific genes are hidden by condensation of some chromosomes.
- 40.** Some plant tissue culture techniques involve the production of protoplasts using leaf tissue. The tissue is treated to obtain isolated, structurally altered cells. The treatment involves incubation in a solution that contains enzymes such as cellulose and pectinase.

Which statement about protoplast is not correct?

- A** Protoplasts are pluripotent and regenerate into whole plants when provided with the correct growth factors.
- B** Protoplasts are more susceptible to microbial contamination.
- C** Protoplasts are used for the production of genetically engineered plants as they take up naked DNA easily.
- D** Protoplasts need to be maintained in the solution of the same water potential to prevent lysis.

End of Paper